CDE4301 Innovation & Design Capstone

Final Assessment (60%)

This assessment evaluates the **entirety** of your project which includes your ability to build a strong case for your project, the extent of your knowledge about the current state-of-art, the soundness and/or novelty of your chosen approaches to address the issues of interest in your project, the quality of work that you have put in to realize these approaches, and your achievement of the intended deliverables of the project. The effort that you put into your project work, as well as the initiative that you show in managing the direction and execution of the project, will also be assessed.

If you are working on a more **design-focused** project, you are expected to perform the following:

- Define the **problem of interest** and establish the **value proposition** for your project through primary and secondary research (e.g. market analysis, user surveys, patent search, etc.).
- Establish the **design specifications** that you are required to meet in order to effectively address the problem.
- Generate, evaluate, and select potential **concept solutions** to address the problem.
- Perform **detail design** of the selected concept solution.
- Create and test **prototypes** of the selected concept solution, and **iterate** if necessary.
- **Evaluate** whether the intended deliverables have been met, and make **recommendations** for future work.

If you are working on a more **research-focused** project, you are expected to perform the following:

- Identify the **research gaps** to address through literature review.
- Establish the **hypotheses** that you plan to explore and **scope** of the research to meet in order to effectively meet the objectives of the research.
- Generate, evaluate, and select potential **research methodologies** (e.g. experimental protocols, numerical tools, etc.) to address the gaps.
- Create the **research design matrix** and research protocols (e.g. number of specimens, experimental procedure, etc.) based on the chosen methodologies.
- Collect and analyze data from **tests**, and **iterate** if necessary.
- Form **conclusions** about the research questions, and make **recommendations** for future work.

If your project involves a mix of both design and research elements, you are expected to perform the tasks that are appropriate to these elements depending on the context and scope of your work.

In a nutshell, at the end of your project you are expected to deliver a **convincing outcome** for your project which should comprise the following:

- Clear and compelling case about the **novelty** and/or **value** of your project.
- Detailed, well-articulated, and appropriate **steps** taken to accomplish the intended deliverables of the project.
- **Evidence** which demonstrates that the intended deliverables of the project have been met successfully.

For this assessment, you are required to provide an **online report/documentation** (website on GitHub) of your project and the work that has been completed over **both semesters**. For **individual** projects, the extent of the report/documentation should not exceed **5,000 words** (excluding appendices). For **group** projects, the limit excluding appendices is **7,000 words** for 2 persons; **9,000 words** for 3 persons; **11,000 words** for 4 persons; and **13,000 words** for 5 persons. You should **indicate your individual areas of work clearly** if you are working in a group project. The **online report/documentation** is due at the end of **Week 11 of Semester 2**.

You are also required to give an oral **presentation** (including prototype demonstration if necessary) of not more than **20 minutes** for **individual** projects or **40 minutes** for **group** projects. The presentation will be held on **Week 13 of Semester 2**.

Your supervisor(s) and examiner will assess the work that you have done based on the content of your thesis and presentation. In addition, your examiner will assess your writing and presentation skills, while your supervisor(s) will assess your effort and initiative from Week 1 to Week 12 of the semester.

A. Project execution and achievement – graded by SUPERVISOR(S) and EXAMINER

Level of achievement	A1. Objective, Scope and Context
Key questions to address	*What problem are you trying to address in your project? What are you trying to achieve? How is your project different from existing or past work?* *What is the setting of the problem you are trying to address? Where is your project situated within the bigger scheme of things?*
Poor	☐ Objective of the project is **undefined**. ☐ Discussion about the **novelty** or **value** of the problem or research gap is almost **non-existent.** ☐ Discussion about the context of the problem is almost **non-existent**.
Emerging	☐ Objective and scope of the project are **unclear**. ☐ Discussion about the novelty or value of the problem or research gap is **superficial,** using **uncompelling** reasons. ☐ Discussion about the context of the problem is **superficial** and/or using information that is mostly **irrelevant** or **outdated**.
Competent	☐ Objective and scope of the project are **clearly** defined. ☐ Design specifications are **clearly presented** and will likely meet the intended deliverables of the project. ☐ **Clear** articulation of the project's **novelty** and/or value proposition. ☐ **Clear** and **broad** discussion about the context of the problem using information gathered from a range of **relevant** and **up-to-date** sources.
Exceptional	☐ Objective and scope of the project are **clearly defined and feasible**. ☐ Design specifications are **clearly presented**, **well justified**, and **coherent** the intended deliverables of the project. ☐ **Compelling** articulation of the project's **novelty** and/or value proposition. ☐ **Comprehensive** and **detailed** discussion about the context of the problem using information gathered from an **extensive** range of **relevant**, **authoritative**, and **up-to-date** sources.

Level of achievement	A2. Methodology / concept development
Key questions to address	*How do you intend to achieve the intended deliverables of your project? How do you know that your approach is suitable?*
Poor	☐ Chosen methodologies/concept solutions are **unclear**.
Emerging	☐ Chosen methodologies/concept solutions are **identified** but evaluation/justification and comparison with alternatives are **missing**.
Competent	☐ Chosen methodologies/conceptual solutions are **clearly identified** and **evaluated/justified** with comparison with alternatives.
Exceptional	☐ Chosen methodologies/concept solutions and their **alternatives** are evaluated in a **systematic** and **comprehensive** manner.

Level of achievement	A3. Fabrication of specimens/prototypes
Key questions to address	*What have you created to achieve the intended deliverables of your project? What are the purposes of these specimens or prototypes? How did you create these specimens or prototypes?*
Poor	☐ **Little or no work done** to execute the project – prototypes/specimens **could have been made, but not carried out.**
Emerging	☐ Prototypes/specimens are selected and/or created but **understanding** of knowledge, principles, and theories in relevant fields is **superficial**. Intended purposes of prototypes/specimens may be **unclear**.
Competent	☐ Prototypes/specimens are selected and/or created with **sound application** of knowledge, principles, and theories in relevant fields. Intended purposes of prototypes/specimens are **clearly communicated**.
Exceptional	☐ **Multiple** prototypes/specimens are **thoughtfully** created with **extensive references** to and **sound application** of knowledge/principles/theories in relevant fields. There is clear evidence of **fine workmanship** in the prototypes/specimens made.

Level of achievement	A4. Testing
Key questions to address	*What tests have you performed to investigate the key questions in your project? What are the purposes of these tests? How did you conduct these tests?*
Poor	☐ **Little or no work done** to execute the project – testing **could have been done, but not carried out**.
Emerging	☐ Intended purposes/scope of testing may be **unclear**, and approaches selected for testing of prototypes/specimens may be **unsuitable** for the intended purposes.
Competent	☐ Intended purposes/scope of testing are **clearly defined**, and testing of prototypes/specimens is conducted **satisfactorily** using approaches that are appropriate for the intended purposes.
Exceptional	☐ Intended purposes/scope of testing are **clearly defined**, and testing of prototypes/specimens is conducted **conscientiously** using the best possible approaches for the intended purposes.

Level of achievement	A5. Analysis
Key questions to address	*What have you found out through the specimens or prototypes which you have created and/or tests which you have conducted in your project? How do you know that your findings are valid?*
Poor	☐ **Little or no** analysis/explanation of data/observations.
Emerging	☐ **Minimal** analysis/explanation of data and observations or using **inappropriate** methods, techniques, theories, and/or concepts.
Competent	☐ Analysis/explanation of **critical** data and observations is performed but use of relevant methods, techniques, theories, and/or concepts to demonstrate or explain key behavior/trends **may not be entirely correct**.
Exceptional	☐ **In-depth** analysis/explanation of data and observations is performed through **precise** use of appropriate methods, techniques, theories, and/or concepts to demonstrate or explain key behavior/trends.

Level of achievement	A6. Recommendations for future work
Key questions to address	*What are the shortcomings in your work? How do you think they should be resolved?*
Poor	☐ **No awareness** of shortcomings of work done. ☐ **No Plan** for any future work.
Emerging	☐ **Some awareness** of shortcomings of work done ☐ **D**oes not have recommendations to resolve these shortcomings.
Competent	☐ Major shortcomings of work done are **identified.** ☐ Recommendations to resolve these shortcomings **may not be clear or viable**.
Exceptional	☐ Major shortcomings of work done are **clearly identified** ☐ **Sound** and **well-justified** recommendations to resolve these shortcomings.

Level of achievement	A7. Impact / Visibility
Key questions to address	*How significant is the impact of your work beyond this project module?*
Poor	☐ The outcomes have **little or no impact** beyond the project module.
Emerging	☐ It is **uncertain** whether the outcomes have any impact beyond the project module.
Competent	☐ The outcomes **could be** publishable, patentable, and/or adopted for real-world application after **significant** further improvements.
Exceptional	☐ Considerable **impact** in the field of work: of **publication standard** or is **patentable**, and/or has excellent potential for further development or **real-world application**, and/or of high **visibility** (e.g. achievement in prestigious competitions, showcased at national or international events).

B. Report/documentation – graded by EXAMINER only

Level of achievement	B1. Structure and organisation
Poor	☐ The report is **incomplete**. Most of the key parts are **missing**. ☐ **Little or no details** provided to support the main points.
Emerging	☐ Transition/sequence of ideas **lacks clarity** and/or is **abrupt** at times, and/or work done tends to be presented in **chronological** rather than logical sequence. ☐ **Many** main points **could be better** supported with appropriate details.
Competent	☐ **Effective** transition/sequence of ideas through organisation of chapters and sections that follows a clear and mostly complete **logical** sequence. ☐ **Most** of the main points are supported with appropriate details.
Exceptional	☐ **Exemplary** writing with **logical** organisation of chapters and sections which gives a **clear**, **precise**, **comprehensive**, and **complete** picture about all the main points covered in the report. ☐ **All** main points are **strongly supported** with **accurate** and appropriate details.

Level of achievement	B2. Use of illustrations
Poor	☐ Illustrations are **disorganized,** indistinct and contain **major errors**.
Emerging	☐ Illustrations **lack visual appeal** and are **ill-chosen** (i.e. not fitting for their intended purposes) ☐ Illustrations lack clarity (e.g., too small and indistinct to see clearly), clear captions, annotations, axis labels (for graphs and charts), units (for numbers) and/or contain **many minor errors**.
Competent	☐ Appropriate use of illustrations to communicate key ideas/information but **could be more clearly annotated**.
Exceptional	☐ **Extensive** use of **properly annotated** and easily understood illustrations to communicate key ideas/information.

Level of achievement	B3. References
Poor	☐ Sources of information are almost **non-existent**.
Emerging	☐ Sources of information are **incomplete** and/or referenced using **incorrect** styles.
Competent	☐ Sources of information are referenced but may contain some **minor errors and inconsistencies**.
Exceptional	☐ Sources of information are **complete** and referenced using appropriate and **consistent** styles.

C. Presentation – graded by EXAMINER only

Level of achievement	C1. Quality of presentation material
Poor	☐ Presentation material is **missing** many key parts, **inconsistent**, disorganized, and/or contains many **major errors**. ☐ **Little or no** use of illustrations to communicate key ideas/information.
Emerging	☐ Presentation material **lacks visual appeal,** and contains many **minor** errors and inconsistencies that are **distracting** and/or may cause some confusion. ☐ Illustrations are **ill-chosen** (i.e. not fitting for their intended purposes). ☐ Illustrations lack clarity (e.g., too small and indistinct to see clearly), clear captions, annotations, axis labels (for graphs and charts), units (for numbers) and/or contains **many minor errors**.
Competent	☐ Presentation material is **visually pleasing** and **well-organized** but may contain **minor** errors and inconsistencies that are not distracting or confusing. ☐ Good use of illustrations to communicate key ideas/information but **could be more clearly annotated**.
Exceptional	☐ Presentation material has **excellent visual appeal**, well-organized and virtually **error-free**, and looks **professionally made**. ☐ **Extensive** use of **properly annotated** and easily understood illustrations to communicate key ideas/information.

Level of achievement	C2. Delivery and presentation skills
Poor	☐ Presentation is **barely comprehensible** and throughout. ☐ Presentation is **awkward** throughout. There is **little evidence** of practice/preparation.
Emerging	☐ Transition/sequence of ideas **lacks clarity** and/or is **abrupt** at times, and/or work done tends to be presented in **chronological** rather than logical sequence. ☐ Some evidence of practice/preparation, but time management **needs to be improved** and/or pace of delivery is too slow or too fast.
Competent	☐ Clear transition/sequence of ideas but **logical flow can be improved** and/or presentation **can be more concise**. ☐ Good time management and pace of delivery. Clear evidence of practice/preparation but can be more confident.
Exceptional	☐ Clear, **comprehensive**, **concise**, and **compelling** presentation with **excellent logical flow**. ☐ **Precise** time management and pace of delivery. **Confident** presentation with clear evidence of practice/preparation

Level of achievement	C3. Response to questions/comments
Poor	☐ **Unable to respond** to questions/comments. **Little evidence** of practice/preparation.
Emerging	☐ Needs **repeated prompting** to respond to questions/comments. ☐ Responses to questions/comments are **superficial** and suggest lack of understanding about the subject matter.
Competent	☐ Responses to questions/comments are generally **accurate** and may sometimes lack clarity and/or could be more concise.
Exceptional	☐ Responses to questions/comments are **thoughtful**, **accurate**, and **concise.** Clear evidence of practice/preparation.

D. Effort and Initiative – graded by SUPERVISOR(S) only

Level of achievement	D1. Motivation
Poor	☐ **Unmotivated** and demonstrates minimal attempt to achieve required tasks. ☐ Frequently makes excuses for not carrying out work.
Emerging	☐ Attempts to achieve required tasks but needs to be **prompted** frequently. ☐ Tends to stop working when difficulties arise
Competent	☐ Well-motivated and demonstrates **consistent** attempts to execute required tasks. ☐ Attempts to continue working when difficulties arise.
Exceptional	☐ Highly motivated and always exerts **maximal** attempts to execute required tasks. ☐ Always **perseveres/persistent** in tackling problems encountered even when solutions may not be immediately obvious.

Level of achievement	D2. Attitude towards project meetings
Poor	☐ Frequently **skips** or **late** for meetings.
Emerging	☐ Punctual for meetings but is **unprepared** for and/or **seldom** contributes during meetings.
Competent	☐ Mostly punctual and **well-prepared** as well as contributes **actively** during meetings.
Exceptional	☐ **Always** punctual and **well-prepared** as well as contributes actively and **thoughtfully** during meetings.

Level of achievement	D3. Independent inquiry
Poor	☐ Demonstrates **no interest** in exploring relevant questions for further inquiry unless instructed. ☐ **Persistent** guidance or help from supervisors is given but **not followed**.
Emerging	☐ Demonstrates **some interest** in exploring relevant questions but may not act on tasks for further inquiry. ☐ Guidance or help from supervisors is **frequently** required but sometimes not followed.
Competent	☐ Generally able to generate relevant questions and execute tasks for further inquiry. ☐ Attempts to learn and overcome problems independently but **sometimes** require guidance or help from supervisors
Exceptional	☐ **Constantly** generates relevant questions and executes tasks for further inquiry. ☐ Able to learn and overcome problems with **minimal** guidance or help from supervisors.

Level of achievement	D4. Ownership and resourcefulness
Poor	☐ Displays **zero ownership** of the project.
Emerging	☐ Ownership of the project mostly by **supervisors**. ☐ Frequently **waits** for supervisors to provide resources for the project.
Competent	☐ **Frequently** attempts to self-manage the direction and execution of the project. ☐ **Actively proposes** resources required for the project to supervisors.
Exceptional	☐ Displays clear **ownership** and **leadership** in self-managing the direction and execution of the project. ☐ **Actively proposes** and **secures** resources for the project on his/her own.